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Department of Computer Technology
Code 23CT1704, Sub. Name: Lab Software Engineering

Experiment No. 3

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Aim :

To study and draw the data flow diagrams at level 0 and level 1 for the given case Study.

Problem Statement:

Design the Data Flow Diagram (DFD) for an E-Commerce Business Management System. The system should manage customer registration, product catalog browsing, shopping cart and order placement, payment processing, shipping and delivery, and generation of sales/delivery reports.

What is a Data Flow Diagram (DFD)?

A **Data Flow Diagram (DFD)** is a graphical representation of how data moves through a system. It shows:

- External entities (users or other systems)
- Processes (activities performed)
- Data stores (files or databases)
- Data flows (movement of data between entities, processes, and data stores)

DFDs are commonly used during the System Analysis and Design (SAD) phase of software development to understand and document system requirements.

Basic DFD Symbols

Symbol	Name	Purpose
Rectangle	External Entity	Source or destination of data (e.g., Customer, Admin)
Circle / Rounded Rectangle	Process	Performs operations on data
Open-ended Rectangle / Parallel Lines	Data Store	Stores data (database or file)
Arrow	Data Flow	Shows movement of data

Level 0 DFD (Context Diagram)

A **Level 0 DFD**, also called the Context Diagram, provides a high-level view of the entire system.

Characteristics

- Represents the whole system as one single process.
- Shows only external entities interacting with the system.
- Does not show internal processes or data stores.
- Used to understand the overall scope of the system.

Purpose

- Shows who interacts with the system.
- Defines system boundaries.
- Gives an overview of data entering and leaving the system.

Level 1 DFD

A **Level 1 DFD** expands the single process shown in Level 0 into multiple detailed processes.

Characteristics

- Breaks the main system into major functions.
- Includes processes, data stores, and data flows.
- Shows how data moves between different parts of the system.
- Provides more detail than the Context Diagram.

Problem Statement

Design the **Data Flow Diagram (DFD)** for an **E-Commerce Business Management System**. The system should manage customer registration, product catalog browsing, shopping cart and order placement, payment processing, shipping and delivery, and generation of sales/delivery reports.

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Case Study: E-Commerce Business Management System

1) Level 0 DFD (Context Diagram)

The entire E-Commerce Business Management System is represented as one single process.

Process

Process No.	Process Name
0.0	E-Commerce Business Management System

External Entities

Entity	Description
Customer	Browses products, registers/logs in, places orders, and receives order confirmations.
Admin / Store Manager	Manages product catalog and receives sales/order reports.
Payment Gateway	Processes payment requests and returns payment confirmation.
Delivery Partner	Receives shipment details and provides delivery status updates.

Data Flows Between Process and Entities

1. Customer ↔ E-Commerce Business Management System

From	To	Data Flow
Customer	E-Commerce Business Management System	Registration Info, Order Details
E-Commerce Business Management System	Customer	Product Info, Order Confirmation, Invoice

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2. Admin / Store Manager ↔ E-Commerce Business Management System

From	To	Data Flow
Admin / Store Manager	E-Commerce Business Management System	Product Details, Stock Updates
E-Commerce Business Management System	Admin / Store Manager	Sales Reports, Order List

3. E-Commerce Business Management System ↔ Payment Gateway

From	To	Data Flow
E-Commerce Business Management System	Payment Gateway	Payment Request
Payment Gateway	E-Commerce Business Management System	Payment Confirmation

4. E-Commerce Business Management System ↔ Delivery Partner

From	To	Data Flow
E-Commerce Business Management System	Delivery Partner	Shipment Details
Delivery Partner	E-Commerce Business Management System	Delivery Status Update

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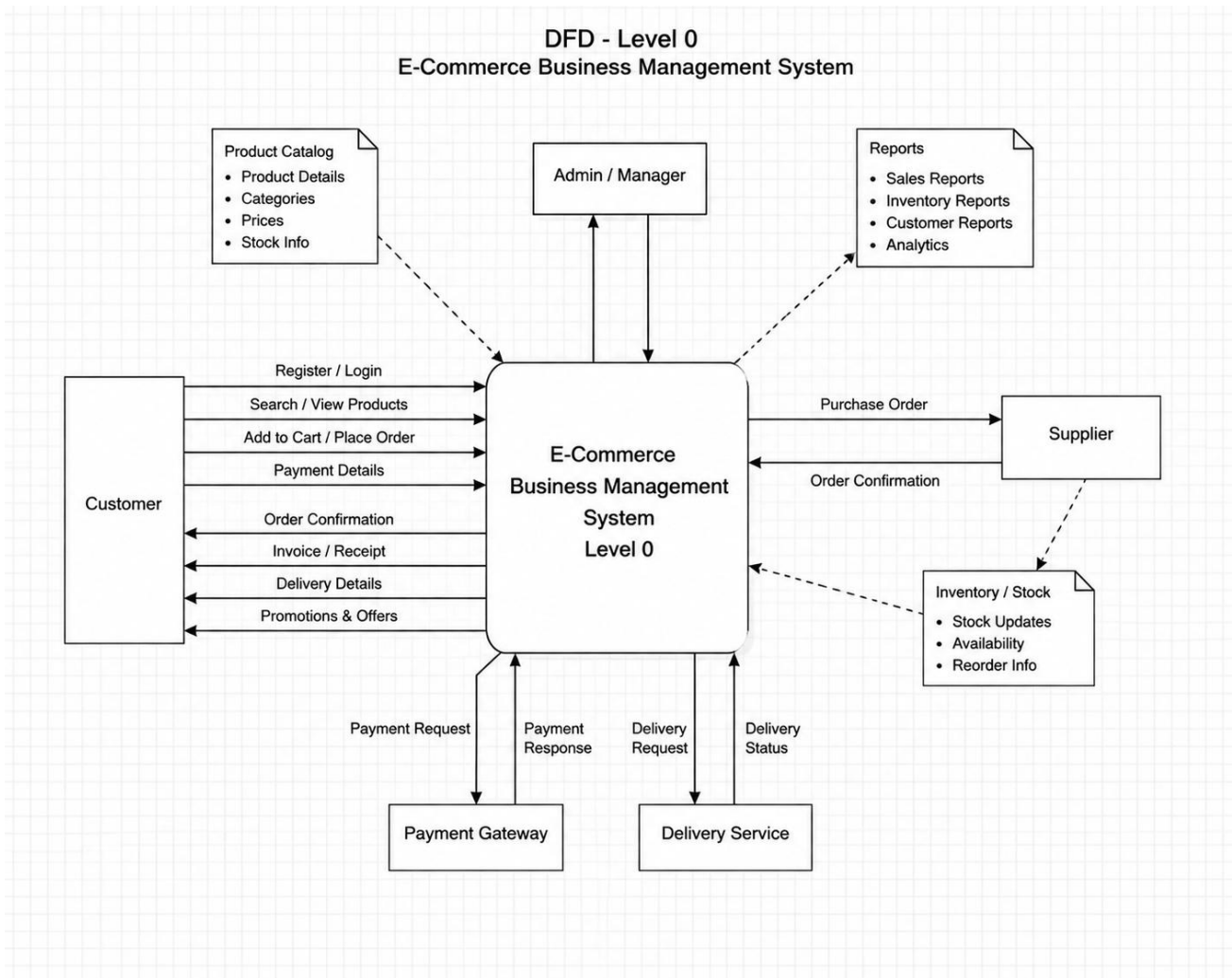
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Level 0 DFD Diagram



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2) Level 1 DFD

The main process is divided into smaller sub-processes.

The E-Commerce Business Management System has 4 processes, 4 external entities, and 5 data stores.

1. Processes

Process No.	Process Name	Description
1.1	Manage Product Catalog	Maintains product listings and updates catalog information.
1.2	Process Order & Cart	Manages shopping cart, order placement, and order records.
1.3	Process Payment	Handles payment requests and communicates with the payment gateway.
1.4	Manage Shipping & Delivery	Coordinates shipment creation, tracking, and delivery reporting.

2. External Entities

Entity	Description
Customer	Provides registration/order info and receives product, order and delivery updates.
Admin / Store Manager	Provides product information and receives sales/delivery reports.
Payment Gateway	Receives payment requests and returns payment confirmation.
Delivery Partner	Receives shipment details and returns delivery status updates.

3. Data Stores

Data Store	Description
D1 - Product Database	Stores product details, pricing, and stock information.
D2 - Order Database	Stores order details and order status.
D3 - Payment Database	Stores payment transaction details.
D4 - Customer Database	Stores customer profile and account information.
D5 - Shipping Database	Stores shipment and delivery tracking details.

4. Data Flow Between Processes and Entities

Process 1.1 - Manage Product Catalog

- External Entities: Customer, Admin / Store Manager
- Input: Product Search (Customer), Add / Update Product Info (Admin)
- Output: Product Info / Catalog List (Customer)
- Data Store Used: D1 - Product Database

Process 1.2 - Process Order & Cart

- External Entity: Customer
- Input: Order Details, Cart Items
- Output: Order Confirmation, Invoice
- Data Stores Used: D2 - Order Database, D4 - Customer Database

Process 1.3 - Process Payment

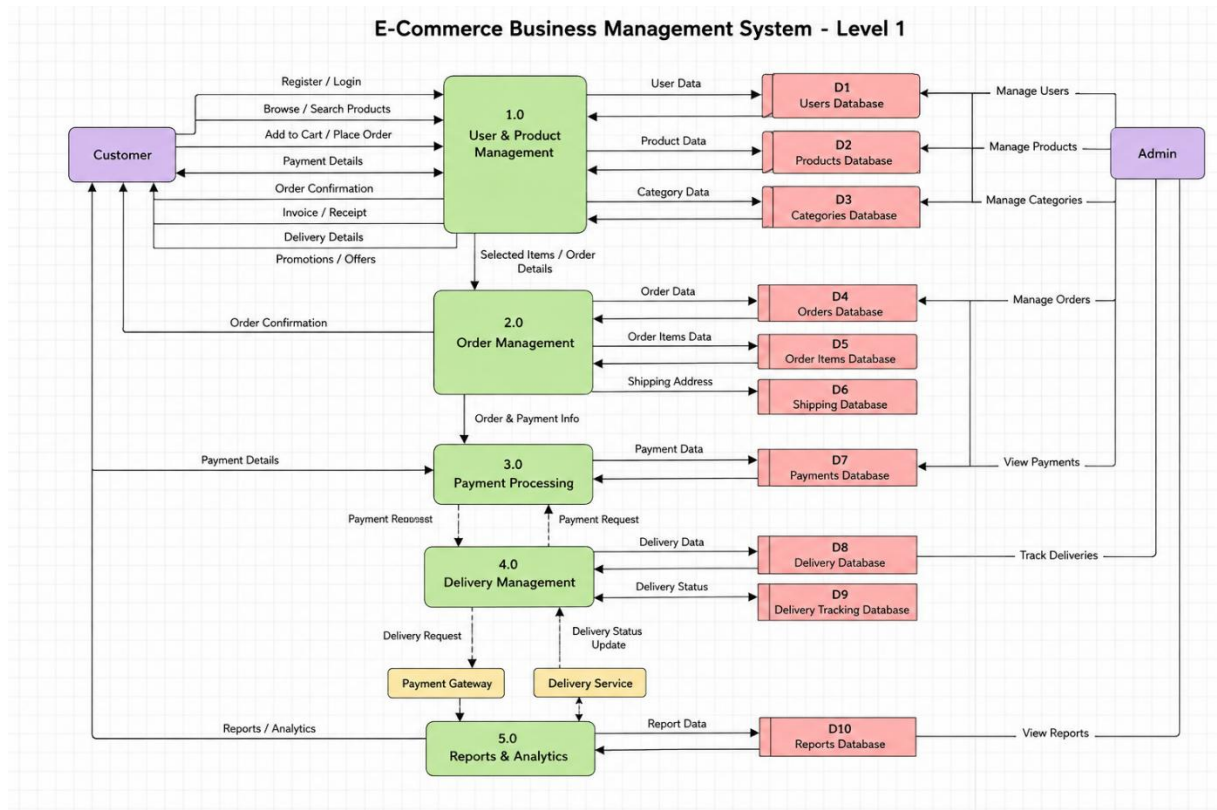
- External Entity: Payment Gateway
- Input: Payment Request (from Order & Cart process)
- Output: Payment Confirmation, Confirmed Order (to Shipping & Delivery)
- Data Store Used: D3 - Payment Database

Process 1.4 - Manage Shipping & Delivery

- External Entities: Delivery Partner, Admin / Store Manager
- Input: Confirmed Order, Delivery Status Update
- Output: Shipment Details (Delivery Partner), Delivery / Sales Report (Admin)
- Data Store Used: D5 - Shipping Database

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Level 1 DFD Diagram



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Steps to Draw the DFD in StarUML

1. Open StarUML.
2. Create a new project.
3. Add a Data Flow Diagram (DFD) (or use basic diagram elements if a DFD extension is unavailable).
4. Draw the Level 0 (Context Diagram): Add one process named E-Commerce Business Management System. Add external entities: Customer, Admin / Store Manager, Payment Gateway, and Delivery Partner. Connect entities to the process with appropriate data flows.
5. Draw the Level 1 DFD: Create processes 1.1 Manage Product Catalog, 1.2 Process Order & Cart, 1.3 Process Payment, 1.4 Manage Shipping & Delivery. Add data stores D1-D5. Connect external entities, processes, and data stores using directional data flows.
6. Verify that all inputs, outputs, and data stores are correctly connected and labeled.

Conclusion

The Data Flow Diagrams (Level 0 and Level 1) for the E-Commerce Business Management System were successfully studied and designed using StarUML. The Level 0 diagram provided a high-level overview of the system boundary and its interaction with external entities, while the Level 1 diagram broke the system down into major processes - Manage Product Catalog, Process Order & Cart, Process Payment, and Manage Shipping & Delivery - along with their associated data stores and data flows, giving a clearer understanding of how data moves through the system.